ORE AMENDMENT APPENDIX

APPENDIX: COMPLETE USER STUDY DATA COLLECTION INSTRUMENTS

- **Study Title:** Evaluating Explainable AI (XAI) Dashboards for Insider Threat Detection
- **Protocol Layout:** Counterbalanced Within-Subject Evaluation Design (N=40)
- **Data Platforms:** Local Embedded Testing Software (Dashboard) & Secure Institutional Qualtrics Environment
- **Date of Revision:** July 2026

PART 1: LOCAL EMBEDDED DASHBOARD INSTRUMENTS

> **Administration Protocol:** These data capture fields are embedded directly within the dashboard's user triage panel. Quantitative decision logs, psychometric workload scales, and open-ended text fields require structural inputs following each of the scenario trials to support local database file compilation and enable the final forensic export option. To fully protect voluntary participation, participants may choose to skip any question by selecting the explicit "Pass" options or by entering "Pass" or "N/A" into text containers prior to file compilation and forensic export.

I. Operational Decision Quality Logs (Categorical Selectors)

1. **Individual Daily Alarm Verdicts** (Prompted 5 times per scenario)
   - **Prompt:** Select Operational Disposition Verdict.
   - **Response Options (Binary Button Choice):**
     - Confirm Threat (Malicious Threat)
     - Dismiss Alarm (Benign Deviation)
2. **Case Closure Classification** (Prompted 1 time per scenario)
   - **Label Prompt:** Case Closure Classification.
   - **Response Options (Dropdown Select Matrix):**
     - S1 - Removable Storage Exfiltration / IP Loss
     - S2 - Active Competitor Solicitation / Corporate Fraud
     - S3 - Host Malicious Misuse via Keylogging Agent
     - S4 - Unauthorized Target Workspace Hijacking
     - S5 - Multi-Vector Overlapping Profile (Active Option for Unsure / Overlapping Indicators)
     - S6 - Dismiss Alarm: Benign User / Normal Anomaly (False Alarm)

    - Pass / Prefer Not to Answer (Ethics Opt-Out)

3. **Decision Certainty Index** (Prompted 1 time per scenario)
   - **Label Prompt:** Analyst Confidence Level.
   - **Response Options (Dropdown Select Matrix):**
     - 5 - Highly Confident (Certainty)
     - 4 - Moderately Confident (High Probability)
     - 3 - Standard Confidence (Indeterminate)
     - 2 - Low Confidence (Speculative)
     - 1 - Skeptical / Highly Uncertain
     - Pass / Prefer Not to Answer (Ethics Opt-Out)
4. **Case Justification Free-Text Field** (Prompted 1 time per scenario)
   - **Label Prompt:** Analyst Case Justification Narrative (3-line entry profile).
   - **Response Constraint:** Open text box entry field.
   - **Placeholder Text:** "Enter tactical diagnostic summary, relevant behavioral updates, and mitigation sign-off rationale details... (If you do not wish to answer, you may enter 'Pass' or 'N/A' to bypass this prompt and proceed.)"

II. Subjective Task Load Metrics (NASA-TLX Short-Form Sliders)

Scale Architecture: 21-point discrete integer slider scale (1 = Low, 11 = Moderate, 21 = High). Interactive tooltips display automatically upon mouse hover to clarify metrics uniformly. Participants who wish to skip a dimension may leave the slider at its default middle setting.

1. **Mental Demand Dimension**
   - **Label Prompt:** Mental Demand (How much mental/perceptual activity was required?)
   - **Hover Tooltip Content:** NASA-TLX: Mental Demand
   - **Definition:** How much mental and perceptual activity was required to complete this triage sequence (e.g., thinking, deciding, calculating, or searching)?
   - **Scale Bounds:** 1 = Straightforward/Simple Context vs. 21 = Extreme Cognitive Complexity/Filtering.
2. **Frustration Level Dimension**
   - **Label Prompt:** Frustration Level (How insecure, discouraged, or annoyed did you feel?)
   - **Hover Tooltip Content:** NASA-TLX: Frustration Level
   - **Definition:** How insecure, discouraged, irritated, stressed, or annoyed did you feel during this specific evaluation trial?
   - **Scale Bounds:** 1 = Smooth and highly satisfying triage vs. 21 = Technical friction and intense analytical stress.

III. Immediate Localized Strategy Tracking (Dynamic Free-Text Fields)

Conditional Software Logic: The dashboard presents only one of the following prompts matching the active interface architecture layout under evaluation.

Frame A Question: Design A Condition (Traditional Baseline Workflow)

- **Prompt Text:** Strategy Feedback: To what extent did the calculated Z-scores in the Feature Table cross-validate or replace your manual searching of the Raw Action Logs?
- **Response Constraint:** Text box container (Minimum 1 sentence required, or explicit entry of 'Pass' or 'N/A' to satisfy structural file validation while opting out).

Frame B Question: Design B Condition (Explainable AI Workflow)

- **Prompt Text:** Strategy Feedback: Did you formulate your thesis directly from the red/blue SHAP waterfall attributions, or did you still feel compelled to open raw logs to verify your decision? Why?
- **Response Constraint:** Text box container (Minimum 1 sentence required, or explicit entry of 'Pass' or 'N/A' to satisfy structural file validation while opting out).

PART 2: GLOBAL RETROSPECTIVE QUALTRICS INSTRUMENTS

> **Administration Protocol:** This independent questionnaire is executed once at the conclusion of Phase 3 (Post-Study Quantitative Metrics) after the participant has experienced both layouts completely. To support voluntary participation, all mandatory "Force Response" configurations have been deactivated and replaced with soft "Request Response" reminders.

I. Demographics and Balanced Cohort Filtering

1. Anonymous Participant Mapping (Analyst_ID)

- **Prompt:** Enter your assigned SOC Analyst ID (1-40).
- **Response Constraint:** Character field with optional numeric integer enforcement bounds.

2. Counterbalancing Order Validation & Sequence Architecture

- **Prompt:** Select your assigned counterbalancing group.
- **Response Options:**
  - Group 1: Evaluated Scenarios 1 & 2 on Design A -> Central Benign (S6) on Design A -> Scenarios 3 & 4 on Design B (Passkey: 12034)
  - Group 2: Evaluated Scenarios 2 & 1 on Design B -> Central Benign (S6) on Design A -> Scenarios 4 & 3 on Design A (Passkey: 21043)
  - Group 3: Evaluated Scenarios 3 & 4 on Design A -> Central Benign (S6) on Design B -> Scenarios 1 & 2 on Design B (Passkey: 34012)
  - Group 4: Evaluated Scenarios 4 & 3 on Design B -> Central Benign (S6) on Design B -> Scenarios 2 & 1 on Design A (Passkey: 43021)

3. Professional Background Cohort (Professional_Background_Cohort)

- **Prompt:** Select your professional background cohort.
- **Response Options:**
  - University Student Cohort (Undergraduate / Graduate Engineering or Computer Science Student)
  - Industry SOC Analyst Cohort (Practicing Professional Cybersecurity / SOC/SecOps Analyst)

II. Retrospective Comparative System Usability Scale (SUS Matrix Table)

Layout Architecture: Side-by-side comparative table layout prompting the participant to grade both dashboard interface experiences concurrently to mitigate contrast bias. Choices track a 5-point Likert Scale: 1 (Strongly Disagree), 2, 3, 4, 5 (Strongly Agree).

- **Design A Assessment** (Comparative_SUS_A)
- **Design B Assessment** (Comparative_SUS_B)

Row Indices (SUS Core Items):

- 1. I think that I would like to use this system frequently.
- 2. I found the system unnecessarily complex.
- 3. I thought the system was easy to use.
- 4. I think that I would need the support of a technical person to be able to use this system.
- 5. I found the various functions in this system were well integrated.
- 6. I thought there was too much inconsistency in this system.
- 7. I would imagine that most people would learn to use this system very quickly.
- 8. I found the system very cumbersome to use.
- 9. I felt very confident using the system.
- 10. I needed to learn a lot of things before I could get going with this system.

III. SOC Dashboard Training & System Satisfaction

- **Satisfaction Scales (Satisfaction_Scales):** Matrix measuring explainability satisfaction, global view interpretation, local view interpretation, and overall ability improvement on a 1 to 5 scale (1 = Very Low / Dissatisfied, 3 = Neutral, 5 = Very High / Satisfied).
- **Evidentiary Justification (Evidentiary_Justification):** Single-answer horizontal multiple-choice question asking whether explainability plots provide clear evidentiary justification for model decisions (Yes/No).
- **Evidence Comments (Evidence_Comments):** Optional essay text box for comments regarding explainability plots as presentation evidence.

IV. Global Qualitative Retrospective Questions (Essay Responses)

- **Qualitative Question 1 (Qual_Q1):** You evaluated identical attack signatures across different timelines using both traditional Z-score tables (Design A) and SHAP visual attribution plots (Design

B). How did your workflow or strategy change when transitioning between these two systems? For instance, did you find yourself looking for different types of evidence?

- **Qualitative Question 2 (Qual_Q2):** When using the interactive Gemini LLM chatbot panel in Design B, did the AI-generated text explanations accurately clarify the local waterfall or global beeswarm distribution profiles? Did interacting with the chatbot alter your confidence level when committing to a final case closure disposition?
- **Qualitative Question 3 (Qual_Q3):** As an analyst, did you feel that the SHAP overlays (waterfall and beeswarm graphs) helped bridge any gaps in specialized cybersecurity domain knowledge, or did they introduce an additional layer of visual complexity that required extra effort to learn?
- **Future Suggestions (Future_Suggestions):** What improvements or additional features would you suggest for this system?